

HW3

Total Points: 100

1. Recall that the heat kernel on $\mathbb{R}^d$ is

$$G_t(x) = \frac{1}{(4\pi t)^{d/2}} e^{-|x|^2/4t}, \quad t > 0.$$

For $\phi \in \mathcal{C}_b(\mathbb{R}^d)$, we define $u(t, x) = (G_t * \phi)(x)$ for $t > 0$ and $u(0, x) = \phi(x)$.

- (a) [5 points] (Semigroup property) Prove directly or using the Fourier transform, that

$$G_t * G_s = G_{t+s}, \quad t, s > 0.$$

- (b) [5 points] Show that

$$\|u(t, \cdot)\|_{L^\infty} \leq \|u(s, \cdot)\|_{L^\infty}, \quad \forall t \geq s \geq 0.$$

[Total for Question 1: 10 points]

2. Consider the non-homogeneous heat equation on $\mathbb{R}^d$

$$\begin{cases} u_t = \Delta u + G_{t+1}(x), & x \in \mathbb{R}^d, t > 0, \\ u(0, x) = 0. \end{cases}$$

- (a) [10 points] Write down the solution using the Duhamel's principle and simplify the solution to

$$u(t, x) = tG_{t+1}(x).$$

Verify the answer by direct differentiation.

- (b) [5 points] Derive the solution if the non-homogeneous term is given by

$$f(t, x) = a(t)G_{t+b}(x), \quad b > 0.$$

[Total for Question 2: 15 points]

3. Consider the heat equation

$$\begin{cases} u_t = u_{xx}, & t > 0, x \in (0, \pi), \\ u(t, 0) = u(t, \pi) = 0, & t > 0, \\ u(0, x) = x(\pi - x), & x \in (0, \pi). \end{cases}$$

- (a) [5 points] Derive the appropriate eigenfunctions and eigenvalues that appears in the separation of variable method.
- (b) [5 points] Compute the Fourier coefficients explicitly and show that

$$u(t, x) = \frac{8}{\pi} \sum_{n \geq 1, n \text{ odd}} \frac{e^{-n^2 t}}{n^3} \sin(nx).$$

- (c) [5 points] What is the leading order behavior of $u(t, x)$ as $t \rightarrow \infty$?
Hint: compute $\lim_{t \rightarrow \infty} e^t u(t, x)$.

[Total for Question 3: 15 points]

4. Consider the heat equation with Neumann boundary condition

$$\begin{cases} u_t = u_{xx}, & t > 0, x \in (0, \ell), \\ u_x(t, 0) = u_x(t, \ell) = 0, & t > 0, \\ u(0, x) = \phi(x), & x \in (0, \ell), \end{cases}$$

where $\phi \in \mathcal{C}^1[0, 1]$ with $\phi'(0) = \phi'(\ell) = 0$.

- (a) [10 points] Prove directly from PDE or using the separation of variable solution, that

$$\frac{1}{\ell} \int_0^\ell u(t, x) dx = \frac{1}{\ell} \int_0^\ell \phi(x) dx, \quad t > 0.$$

- (b) [5 points] Show that

$$u(t, \cdot) \rightarrow \frac{1}{\ell} \int_0^\ell \phi(x) dx, \quad t \rightarrow \infty,$$

uniformly for $x \in (0, \ell)$.

[Total for Question 4: 15 points]

5. [10 points] Use separation of variables and Duhamel's principle to find the classical solution to the PDE

$$\begin{cases} u_t = u_{xx} + e^{-4t} \sin(2x) + \sin(3x), & t > 0, x \in (0, \pi), \\ u(t, 0) = u(t, \pi) = 0, & t > 0, \\ u(0, x) = \sin x, & x \in (0, \pi). \end{cases}$$

[Total for Question 5: 10 points]

6. [10 points] Recall the Green's function on $(0, \ell)$ with Dirichlet boundary condition is

$$G(t; x, y) = \frac{2}{\ell} \sum_{n=1}^{\infty} e^{-(n\pi/\ell)^2 t} \sin \frac{n\pi x}{\ell} \sin \frac{n\pi y}{\ell}, \quad t > 0.$$

Show the semi-group property

$$\int_0^\ell G(t; x, z) G(s, z, y) dz = G(t + s; x, y), \quad t, s > 0.$$

[Total for Question 6: 10 points]

7. Let $G_t(x)$ be the heat kernel in one dimension. For $x, y \in (0, \ell)$, define

$$\tilde{G}(t; x, y) = \sum_{k \in \mathbb{Z}} [G_t(x - y + 2k\ell) - G_t(x + y + 2k\ell)].$$

- (a) [5 points] Show that $(\partial_t - \partial_{xx})\tilde{G} = 0$ and $\tilde{G}(t; 0, y) = \tilde{G}(t; \ell, y) = 0$ for $t > 0$.
(b) [5 points] Show that for any $\phi \in \mathcal{C}[0, \ell]$ with $\phi(0) = \phi(\ell) = 0$,

$$\lim_{t \rightarrow 0+} \int_0^\ell \tilde{G}(t; x, y) \phi(y) dy = \phi(x).$$

*Hint: you can extend ϕ oddly to $[-\ell, \ell]$ and periodically to ϕ_1 on $\mathbb{R}$ with period 2ℓ , and use the identity $\langle \tilde{G}, \phi \rangle = G_t * \phi_1$.*

Remark: as a consequence, $\tilde{G} = G$ is the Green's function by uniqueness of the heat equation solution. Can you prove this directly?

[Total for Question 7: 10 points]

8. Let u be a classical solution of

$$\begin{cases} u_t = u_{xx}, & t > 0, x \in (0, \ell), \\ u(t, 0) = u(t, \ell) = 0, & t > 0, \\ u(0, x) = \phi(x), & x \in (0, \ell), \end{cases}$$

where $\phi \in \mathcal{C}[0, \ell]$ and $\phi(0) = \phi(\ell) = 0$.

- (a) [5 points] Assume first that for constant C ,

$$|\phi(x)| \leq C \sin \frac{\pi x}{\ell}.$$

Without using Fourier series or Green's function, prove that

$$|u(t, x)| \leq C e^{-\pi^2 t / \ell^2} \sin \frac{\pi x}{\ell}, \quad t > 0, x \in (0, \ell).$$

Hint: $v(t, x) = C e^{-\pi^2 t / \ell^2} \sin \frac{\pi x}{\ell}$ is a solution to the heat equation. Use maximum principle on $\pm u - v$.

- (b) [5 points] Assume that $\phi \in \mathcal{C}^1[0, \ell]$ with $\phi(0) = \phi(\ell) = 0$. Show that there is a constant $C > 0$ such that

$$|\phi(x)| \leq C \sin \frac{\pi x}{\ell}.$$

- (c) [5 points] Assume $\phi \geq 0$. Extend ϕ by zero to $\mathbb{R}$ and let

$$U(t, x) = (G_t * \phi)(x).$$

Use the maximum principle to show that

$$0 \leq u(t, x) \leq U(t, x), \quad t > 0, x \in (0, \ell).$$

That is, the Dirichlet boundary condition makes the solution smaller.

[Total for Question 8: 15 points]